

Vinesh Reddy Kankanalapally

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Technical Skills

Languages: Java, Python, C, C++, JavaScript, SQL

Backend & Distributed Systems: Spring Boot, Spring Framework, Spring Security, REST APIs, SOAP, Microservices, Kafka, Hibernate, JPA, Redis, PostgreSQL, MySQL, MongoDB, DynamoDB, NoSQL, JDBC, gRPC, Protocol Buffers, Elasticsearch, Distributed Systems, Event-Driven Architecture

CS Fundamentals: Data Structures, Algorithms, System Design, Object-Oriented Design, Design Patterns, Concurrency, Multithreading, Complexity Analysis, Performance Optimization, API Design, High Availability, Fault Tolerance

AI & Machine Learning: TensorFlow, Keras, Scikit-learn, ANN, CNN-LSTM, Random Forest, K-Means, NumPy, Pandas, Matplotlib, Data Preprocessing, Model Evaluation

Frontend: React.js, HTML, CSS

Cloud & DevOps: Cloud Computing, AWS (EC2, S3, RDS, Lambda, EKS, IAM, CloudFormation, VPC), Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Maven, Gradle, CI/CD

Testing & Tooling: JUnit, Mockito, SonarQube, Postman, Git, Linux, GitHub Copilot, Agile, Scrum, SDLC, Code Reviews, Debugging, Root Cause Analysis

Observability: OpenTelemetry, Prometheus, Grafana, Distributed Tracing, Performance Profiling

Work Experience

Software Engineer – Client: Ford Motor Company

Aug 2024 – Present

Codelark LLC

Dallas, Texas – Remote

- Designed, developed, and maintained large-scale, high-availability, high-throughput backend services using Java, Spring Boot, REST APIs, and microservices for enterprise automotive cloud-native applications.
- Improved backend performance by **30%** by profiling service and SQL bottlenecks, optimizing algorithms, data access patterns, and refactoring latency-sensitive application logic.
- Reduced release-cycle time by **40%** by implementing automated CI/CD pipelines with GitHub Actions, Docker, Kubernetes, Terraform, and SonarQube quality gates.
- Owned features end-to-end across architecture, implementation, code review, unit and integration testing, CI/CD deployment, production monitoring, and root-cause analysis.
- Improved system reliability by deploying containerized microservices on Kubernetes/EKS with AWS, Terraform IaC, and observability via OpenTelemetry, Prometheus, and Grafana.
- Collaborated with product, architects, QA, and DevOps teams to translate requirements into technical designs; reviewed pull requests enforcing correctness, performance, and testing standards.

Software Developer

Sep 2021 – Jul 2022

Add On Systems Information Technology Services

Bangalore, India

- Built and maintained Java, Spring, Hibernate, and REST-based backend components for banking applications, delivering production enhancements and customer-facing functionality.
- Reduced database latency by **15%** by optimizing SQL queries, persistence logic, and data access patterns, achieving query execution under **50 ms** and response times under **100 ms**.
- Applied data structures, object-oriented design, reusable design patterns, and modular architecture to improve backend maintainability and application performance.
- Performed code reviews, defect analysis, unit and regression testing, API documentation, and production troubleshooting in collaboration with analysts and QA engineers.

Projects

Intelligent Cloud Operations Platform | *Java, Spring Boot, Kafka, Kubernetes, PostgreSQL, OpenTelemetry* | [GitHub](#) 2026

- Designed a distributed cloud-operations platform using independently deployable Spring Boot microservices, Kafka event streams, PostgreSQL, and Kubernetes with auto-scaling and fault tolerance.
- Implemented anomaly detection, automated remediation workflows, Kubernetes custom resources, and end-to-end observability with OpenTelemetry, Prometheus, Grafana, Loki, and Tempo.

Banking REST API System | *Java, Spring Boot, Kafka, Redis, MySQL, Docker, Kubernetes* | [GitHub](#)

2023

- Engineered scalable banking services with REST APIs, Spring Security, JWT, Kafka event processing, Redis caching, and Hibernate/JPA; reduced query response time by **15%** via SQL optimization.
- Implemented unit and integration testing, exception handling, API documentation, Docker packaging, and Kubernetes deployment with automated CI/CD pipeline via GitHub Actions.

Real-Time DDoS Attack Detection | *Python, FastAPI, TensorFlow, Scikit-learn, AWS, Kubernetes* | [GitHub](#)

2023

- Built a real-time ML detection service using ANN, CNN-LSTM, Random Forest, and K-Means ensemble models with FastAPI inference endpoints, SHAP explainability, and AWS/Kubernetes deployment, reducing simulated attack impact by **30%**.

Education

University of Colorado Denver

Aug 2022 – May 2024

M.S. in Computer Science, GPA: 3.65/4.00

Denver, Colorado

Vasavi College of Engineering

Jul 2017 – Aug 2021

B.E. in Electronics and Communication Engineering, GPA: 3.50/4.00

Hyderabad, India

Certifications

AWS Certified Solutions Architect – Associate (Jun 2024, Expires 2027) | AWS Certified Developer – Associate (May 2024, Expires 2027) | Microsoft Certified: Azure Fundamentals (Apr 2024)